

GUNNARN, Sweden

WMO: 021260

Lat: 65.0094N

Lon: 17.6993E

Elev: 251

StdP: 98.35

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.7	-27.7	-33.3	0.2	-30.7	-30.1	0.2	-27.6	11.6	0.8	9.2	1.3	0.5	320	0.680

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.9	24.8	15.7	22.9	15.0	20.8	14.2	17.2	22.1	16.1	20.7	15.1	19.3	2.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.4	11.3	18.3	14.4	10.5	17.3	13.3	9.8	16.4	49.1	22.0	46.0	20.8	43.2	19.3	21.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.1	7.3	6.2	-34.3	27.1	2.8	2.0	-36.3	28.6	-37.9	29.8	-39.5	30.9	-41.5	32.4		
(5)			-34.2	18.8	2.7	1.3	-36.2	19.7	-37.8	20.5	-39.3	21.2	-41.3	22.2		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	1.3	-10.9	-10.3	-5.7	0.6	6.4	11.5	14.5	12.6	7.9	1.5	-4.9	-8.7		
	DBStd	10.35	7.79	8.09	5.86	3.72	3.72	3.42	2.82	3.15	3.26	4.00	6.19	7.65		
	HDD10.0	3510	647	569	488	282	124	21	1	10	79	262	447	579		
	HDD18.3	6234	906	803	746	532	371	207	123	178	313	521	697	837		
	CDD10.0	325	0	0	0	0	12	66	141	91	16	0	0	0		
	CDD18.3	7	0	0	0	0	0	1	4	2	0	0	0	0		
	CDH23.3	124	0	0	0	0	4	32	61	26	0	0	0	0		
	CDH26.7	9	0	0	0	0	0	2	6	1	0	0	0	0		
(14) Wind	WSAvg	2.3	1.9	2.0	2.6	2.6	2.7	2.8	2.3	2.0	2.3	2.2	1.8	2.0		
(15) Precipitation	PrecAvg	597	43	32	32	29	40	70	96	83	39	50	45	36		
	PrecMax	863	90	85	54	73	77	116	166	173	69	87	102	80		
	PrecMin	451	7	7	8	6	6	11	28	29	10	18	13	7		
	PrecStd	126	19	22	14	19	20	34	41	56	19	22	29	18		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.8	6.1	8.0	15.8	23.3	26.8	27.4	25.9	19.6	12.3	9.0	5.9		
		MCWB	3.5	3.6	3.8	8.1	13.5	15.7	17.3	17.0	13.8	9.2	7.4	4.1		
	2%	DB	3.6	4.4	5.6	11.4	20.0	23.7	25.0	23.7	16.9	10.3	5.7	4.0		
		MCWB	1.9	2.2	2.7	5.9	11.7	14.8	16.1	15.7	12.6	8.4	4.3	2.3		
	5%	DB	1.6	2.2	4.2	9.0	17.0	21.0	23.1	21.2	14.9	9.2	4.0	2.5		
		MCWB	0.4	0.4	1.7	4.5	10.3	13.3	15.3	15.1	11.5	7.6	2.8	1.3		
	10%	DB	-0.2	0.4	2.7	6.9	14.0	18.7	21.2	19.0	13.5	7.8	2.4	1.0		
		MCWB	-0.9	-0.7	0.4	3.2	8.8	12.2	14.7	14.0	10.7	6.4	1.6	0.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.6	4.3	8.3	14.4	17.4	19.1	18.1	15.1	10.1	7.2	4.2		
		MCDB	5.6	5.7	7.1	14.5	20.9	24.3	23.4	22.4	18.0	11.2	8.7	5.6		
	2%	WB	2.0	2.4	3.0	6.3	12.8	15.6	17.5	16.8	13.4	9.0	4.6	2.5		
		MCDB	3.5	4.3	5.4	10.5	18.4	21.5	22.3	21.4	15.6	10.1	5.6	3.7		
	5%	WB	0.4	0.7	1.8	4.8	10.9	14.3	16.6	15.7	12.2	7.7	2.9	1.3		
		MCDB	1.4	1.9	3.9	8.5	16.3	19.1	21.2	19.9	14.3	9.0	3.7	2.2		
	10%	WB	-1.0	-0.6	0.6	3.5	9.1	13.0	15.5	14.8	11.2	6.4	1.7	0.3		
		MCDB	-0.3	0.4	2.4	6.5	13.2	17.3	19.7	18.5	13.0	7.6	2.3	1.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.7	9.7	10.8	9.5	10.3	9.6	9.9	9.7	8.5	6.4	6.3	8.0		
		MCDBR	8.9	8.4	10.2	13.6	16.3	13.8	13.7	13.7	11.0	8.0	6.0	8.4		
		MCWBR	7.9	7.2	7.9	8.4	9.0	6.2	6.0	6.7	7.1	6.2	5.3	7.5		
		MCDBR	8.2	8.0	9.4	12.0	15.2	12.0	11.9	11.4	9.5	7.5	5.6	7.9		
	5% WB	MCWBR	7.4	7.1	7.5	7.6	8.8	6.0	5.9	6.2	6.6	6.1	5.0	7.2		
(40) Clear-Sky Solar Irradiance	taub		0.188	0.262	0.294	0.327	0.335	0.336	0.348	0.334	0.306	0.287	0.339	N/A		
	taud		1.952	2.175	2.235	2.326	2.470	2.498	2.477	2.505	2.569	2.433	2.720	N/A		
	Ebn at Noon		336	634	783	844	875	881	856	836	791	622	325	N/A		
	Edh at Noon		22	55	83	97	93	93	92	81	62	44	20	N/A		
(44) All-Sky Solar Radiation	RadAvg		0.12	0.68	2.10	3.84	4.73	5.22	4.77	3.58	2.03	0.81	0.19	0.04		
	RadStd		0.01	0.11	0.19	0.28	0.53	0.40	0.53	0.38	0.26	0.13	0.03	0.00		

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (39 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page